

UOW ECON251
Lecture 02
Economic Growth Models
and Sectors



ECONOMIC Growth

1. Introduction



1.1 Key recap

- **Industry** : refers to the **production** of goods and services
- These are measured and presented as a country's **GROSS DOMESTIC PRODUCT (GDP)** *or more precisely* **Real GDP (RGDP)**
- Then compared over time or amongst other economies using actual or in terms of the Economic growth rate



1.2 Economic/GDP Growth

- Defined as the change in real GDP over a period, usually a year

Months – usually quarterly or years

Economic Growth Rate

=

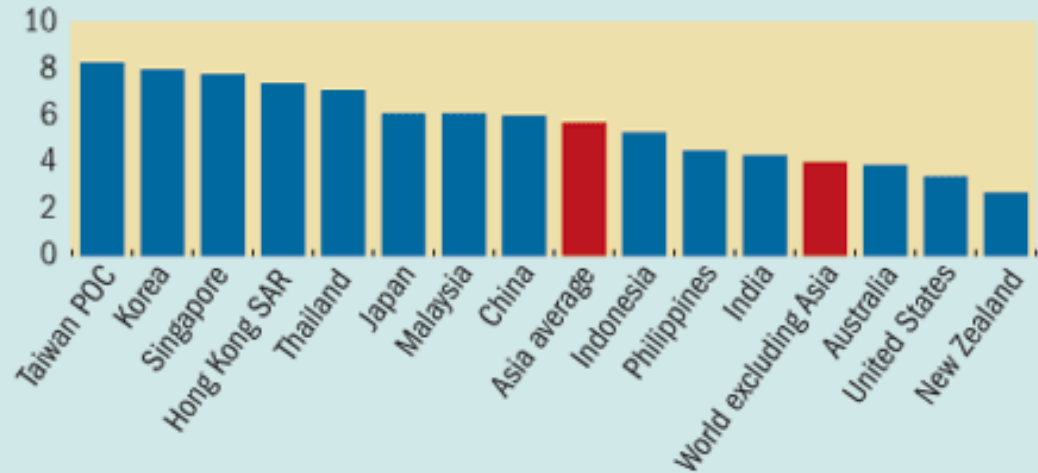
$$\frac{\text{Current Period RGDP} - \text{Previous Period RGDP}}{\text{Previous Period RGDP}} \times 100\%$$

Chart 1

Fast growers

Between 1950 and 1997, Asian countries grew almost twice as rapidly as the average for the rest of the world.

(real GDP growth, average percentage change 1950-97)



Source: Angus Maddison, 2003, *The World Economy: Historical Statistics* (Paris: OECD).

1.3 Computing and interpreting growth rate

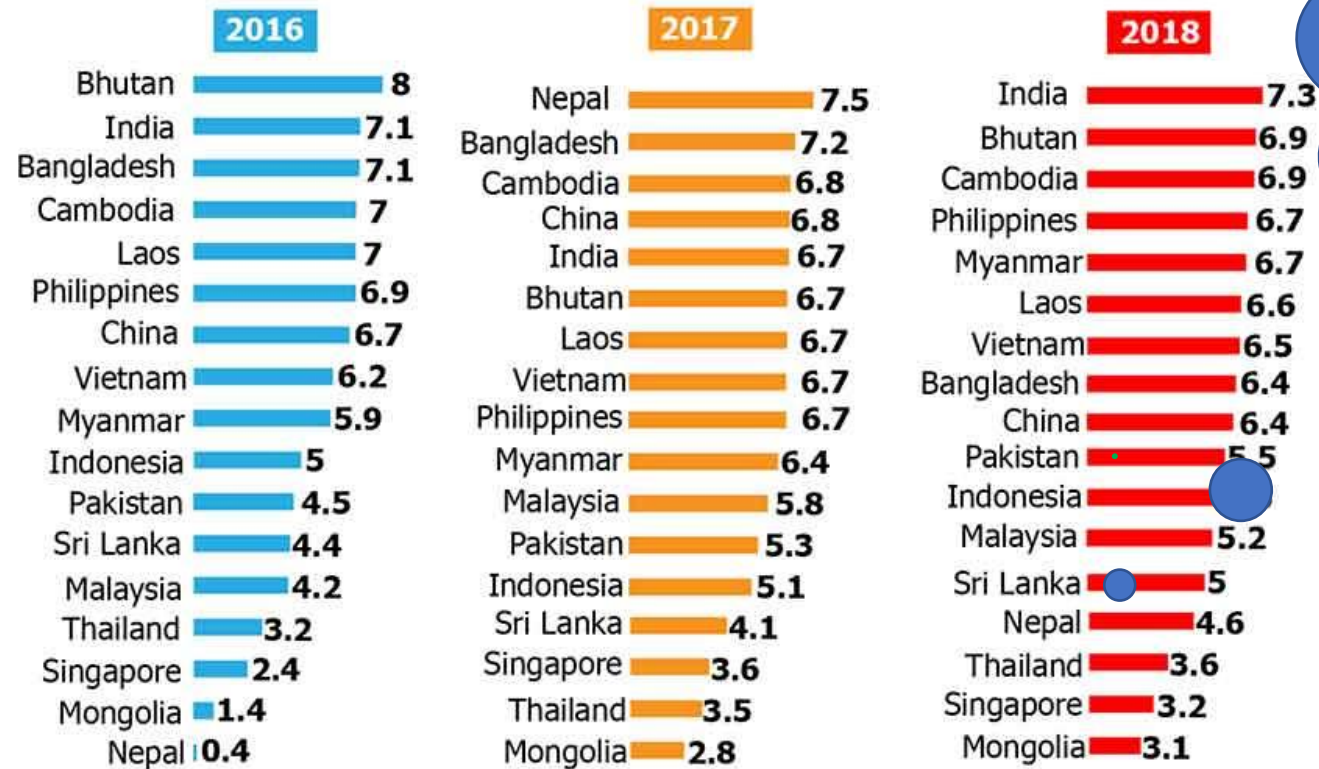
Table 1.1.1 Computing and interpreting economic growth rate

Year	Real GDP (RGDP)	Economic Growth Rate $= \frac{\text{Current Period RGDP} - \text{Previous Period RGDP}}{\text{Previous Period RGDP}} \times 100\%$
2015	800	-
2016	1150	
2017	1600	
2018		0 % [is the economy really stagnating ? What is our reference period?
2019		- 10% [is the economy really poorer ; if we compare eg to Nepal with RGDP = 600 with 22% growth rate}

A quick overview of Asia economies

GDP Growth Projection For Asia, 2018

India is expected to be the fastest growing economy in Asia at 7.3 in 2018, according to the World Bank.



Singapore supposedly one of the most robust economies in Asia but her growth rate was one of the lowest ?

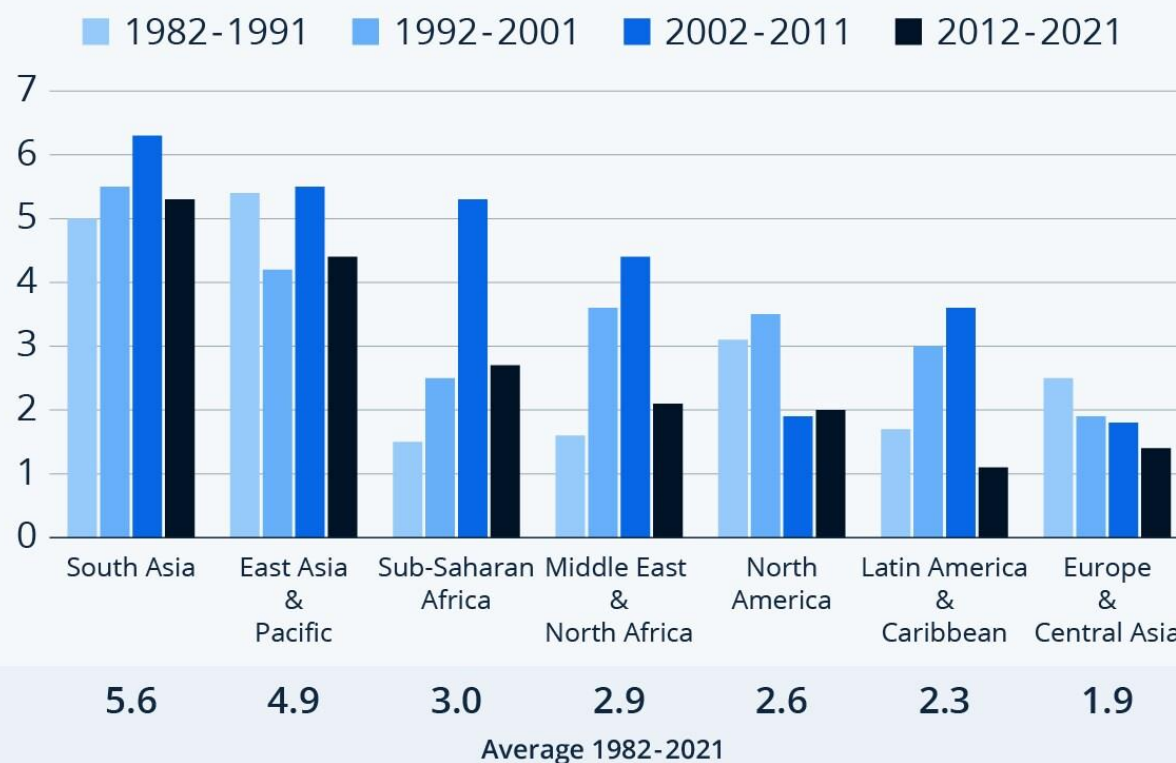
[illegible]

- Japan has been in recession for more than 20 years but she is still the world's top 5 economies.
- How can this be possible ?
- From late 1980s onwards
- Video Link
- <https://youtu.be/wykaDgXoajc>

What is then considered a “decent” economic growth rate ?
What other evidence could we possibly use to further justify the economy is healthy ?

How Global GDP Has Evolved

Average annual GDP growth rate over the periods indicated, by region (in percent)



Source: World Bank

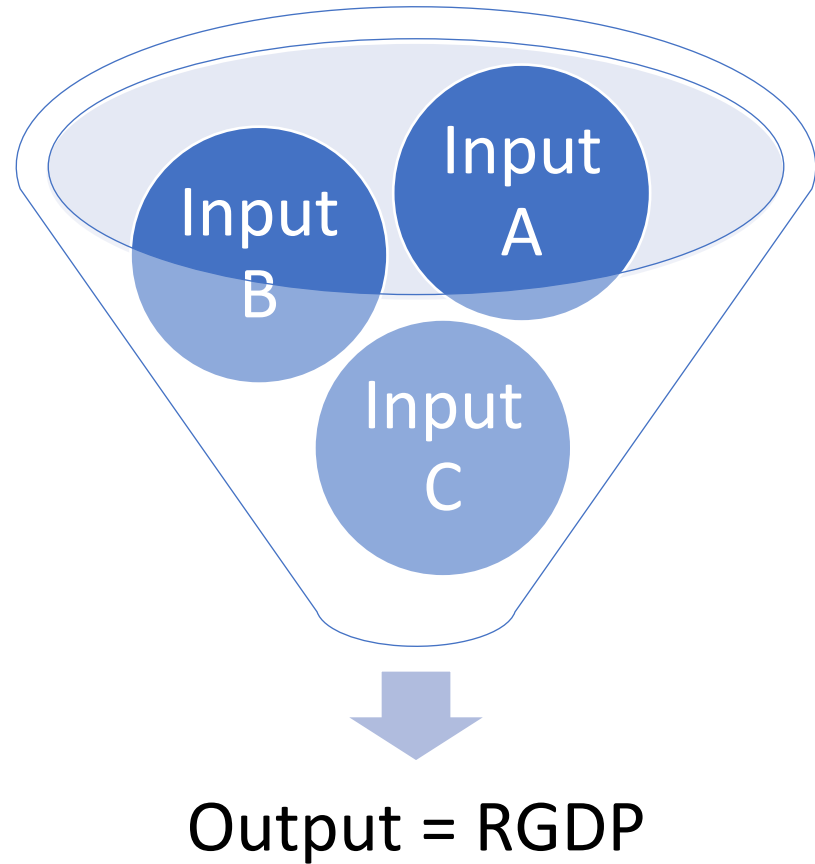


statista

Economic Growth Models



Quick Recap



2.1 GDP Growth – A basic model

Basic model

$$Y = A F(K, L, H, N)$$

Where

Y : Real GDP

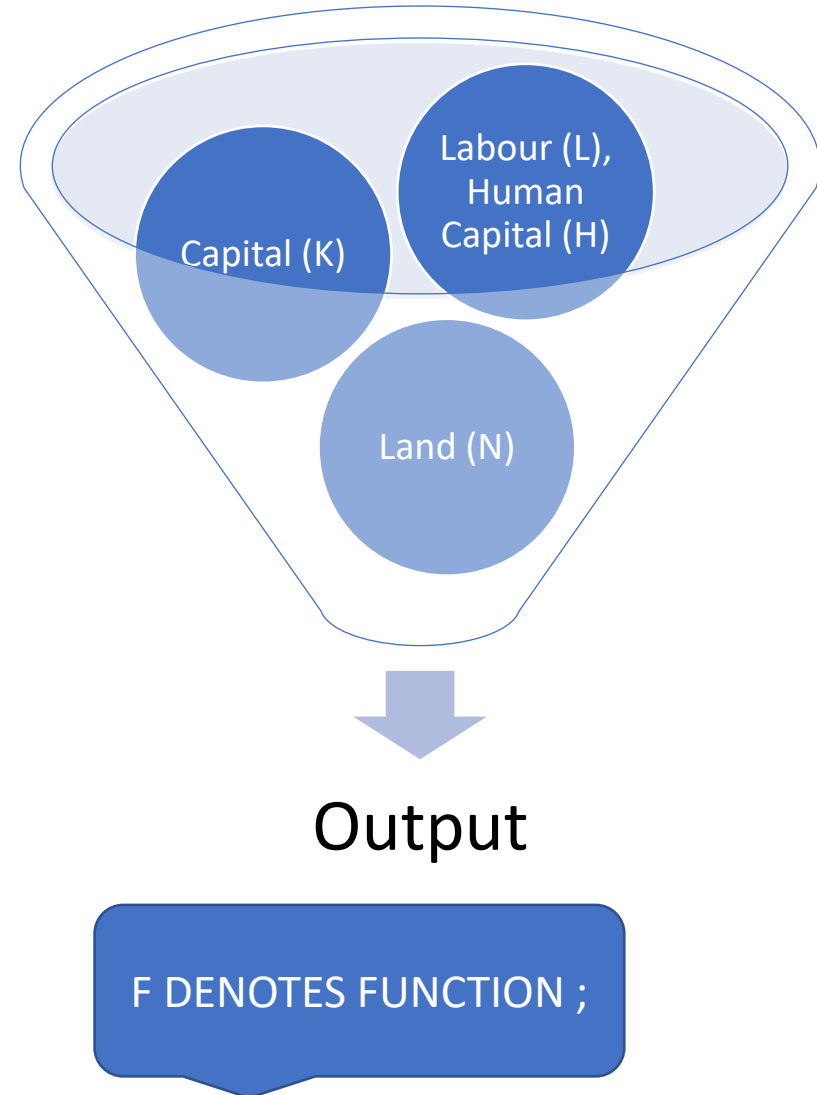
A : Level of technology [This is an **EXOGENEOUS COMPONENT – IE CANNOT BE CONTROLLED BY THE GOVT**]

K : Quantity of capital [Equipment, machines, plants]

L : Quantity of labour

H : Human capital [Knowledge and experience]

N : Land/ Natural Resources



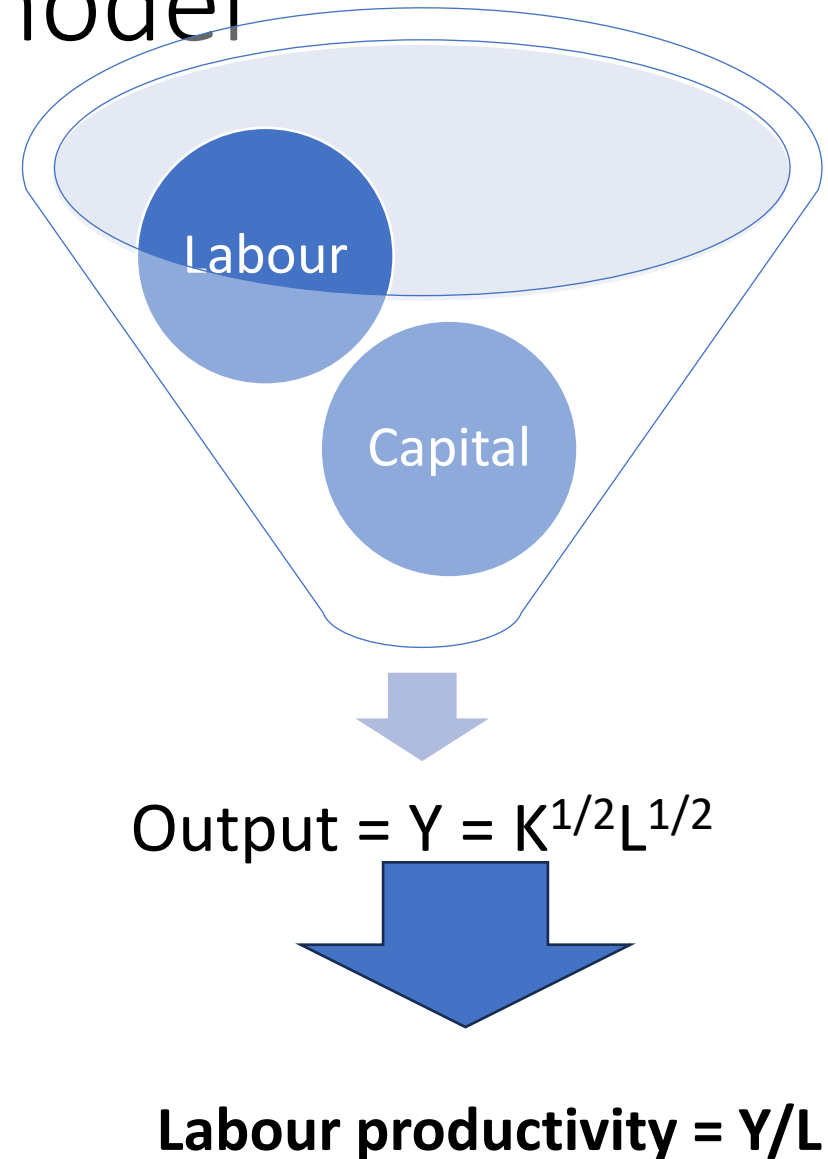
2.1 GDP Growth – A basic model

In most cases, we discuss the role of K and L using an expression the **Cobb Douglas` production function**

$$Y = A(K)^{1/2}(L)^{1/2}$$

Assuming $K = 1$, compute the level of Y given each level of inputs as they both doubled over time

K	L	Y	Y/L
4	4		
8	8		
16	16		

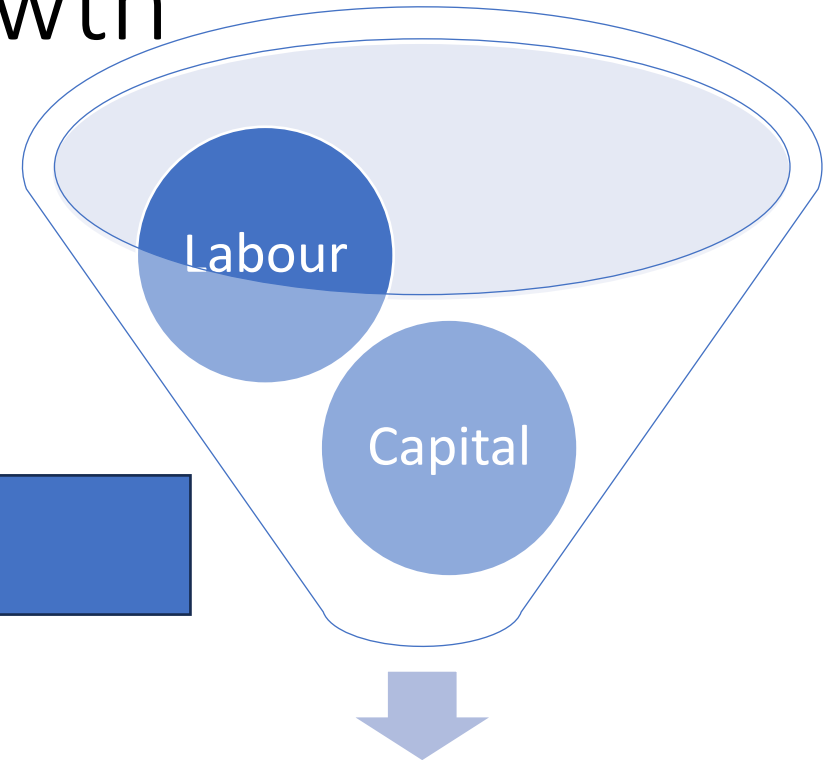


Evidence and causes of growth

Labour force size vs
population
Employment vs
Unemployment rate



Capital Accumulation
Gross Fixed capital formation



Output = RGDP = Y

In the real world (which you will explore real data of assigned Asia economies), **K and L doesn't increase together** as the same rate which affects Y and Y/L

So we **apply linear and multiple regression** for a deeper understanding of which inputs matters more. These will be carried in our **TUTORIAL 2** and group work discussions



Law of Diminishing Returns

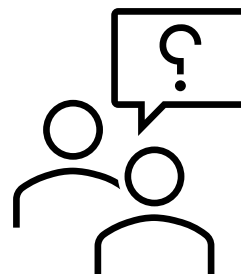
2.1.1 Achieving Growth in the short run

- In the short run, when we can only increase **any one of the inputs** (K or L), Y will initially increase
- BUT the rise in Y gets *Faster / Slower_as K or L rises beyond certain level

As such we say the economy is facing ***Diminishing returns***

Video link :

<https://www.investopedia.com/terms/l/lawofdiminishingmarginalreturn.asp>



2.2.1 Diminishing returns

- **Examine the production function**

$$Y = A(K)^{1/2}(L)^{1/2}$$

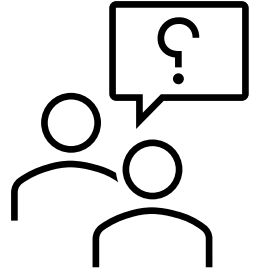
- Assuming A and K constant at 1 and 4 respectively and the country faces a rise in labour force (L)

This is equivalent to countries with large and rising population

So a larger labour force is competing for the use of the fixed level of capital (equipment / machines) , this reduces the **productivity** of the newly added labour

K	L	Y	% Change in Y	Y/L
4	4	4	-	1
4	8	5.66	= [(5.66-4)/4 *100%]= 41.4	0.70
4	12	6.93	=22.4	0.577
4	16	8	=15.4	0.5
4	20	8.94	=11.8	0.447

2.2.1 Diminishing returns



- **Examine the production function**

$$Y = A(K)^{1/2}(L)^{1/2}$$

- Assuming A and K constant at 1 and 4 respectively and the country faces a rise in labour force (L)

This is equivalent to countries with large and rising population

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Diagram :

Diminishing returns

- **Examine the production function**

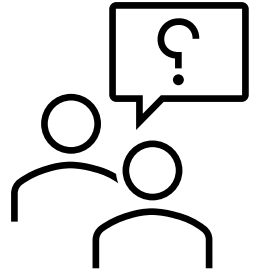
$$Y = A(K)^{1/2}(L)^{1/2}$$

Mathematical proof

Applying first order differentiation

$dY/dK =$

ie as L rises, the rate of increase in Y due to rise in L, falls



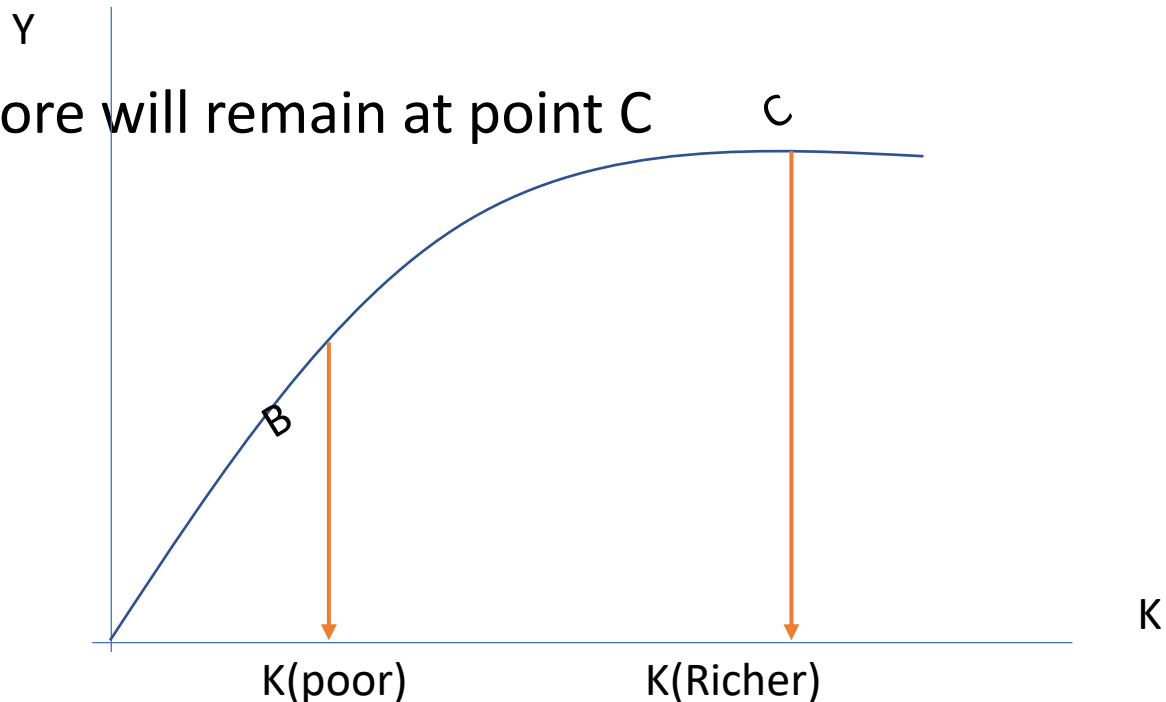
Economic growth and role of capital (k)

- In most economies , govt focus on **the role of capital** (K) rather than the continuous use of labour.
- Thus we will examine the key economic models of economic growth involving the role of capital
- We will revisit the discussion of the role of k in the later topic : Foreign direct investment



2.2 Solow's Growth Model

- Solow's Growth model : Focus on capital accumulation to an ideal level and thereafter any further increase in k will **experience diminishing returns**
- Suggests the poor economies (e.g. Nepal) will be able to catch up with the rich economies (e.g. Singapore) :
- ie moving from point B towards C: international organisations can lend money to Nepal to acquire k
- The assumption here is Singapore will remain at point C



2.3 Solow's Growth Model

Long term growth therefore requires the economy to **apply technology** to given level of labour and capital ratio ; i.e. growth is exogenous – external driven

- Diagram

We observe that richer countries will engage in R&D

This means technology improvement occurs (A rises)

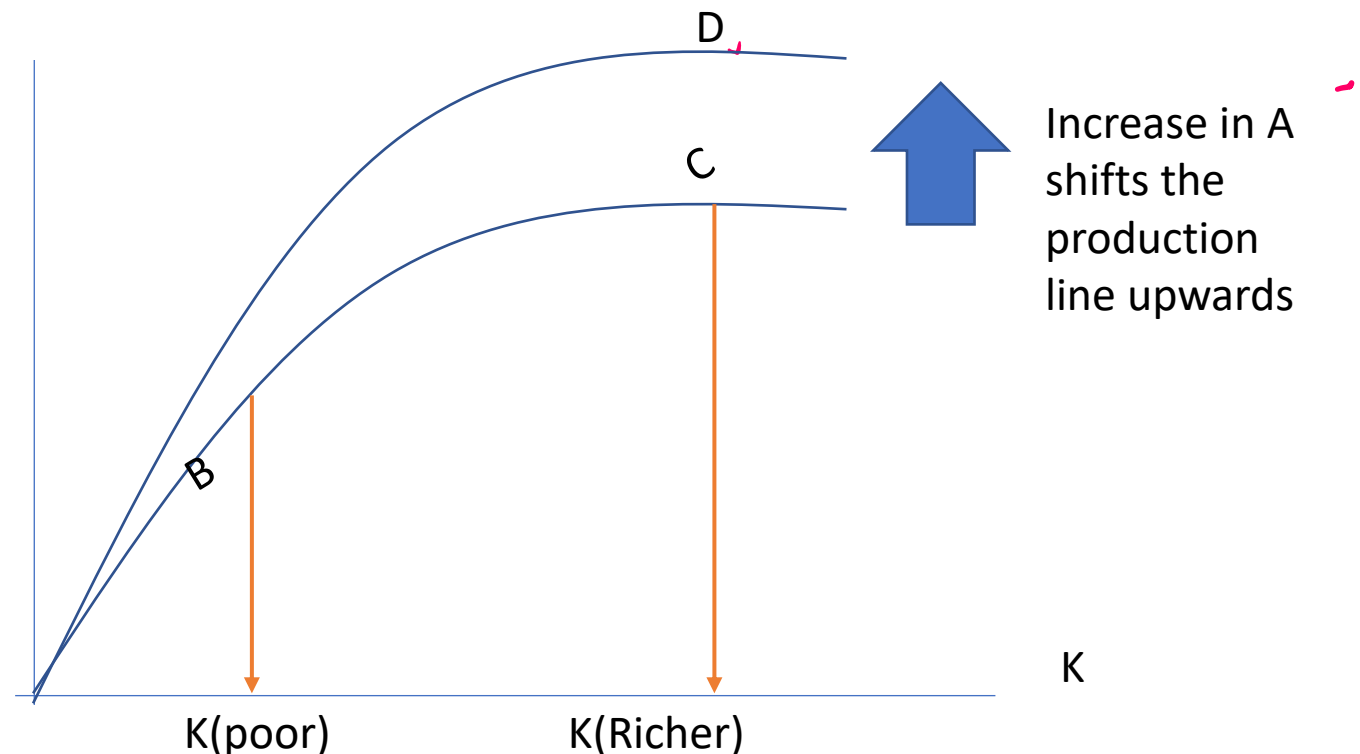
So they move from C to D

So even if poor country increase k, their Y is still lower

- What is the role of govt ?

Promote R&D as k rises but it is expensive and risky

- So we will again examine the role of FDI to overcome this shortfall





Overcoming the odds ???

- So under what circumstances we do not experience diminishing returns or we delay the occurrence ?
- Use either the diagram or consider the mathematical equation

2.3 The AK Model

Now consider this production function

$$Y = AK$$

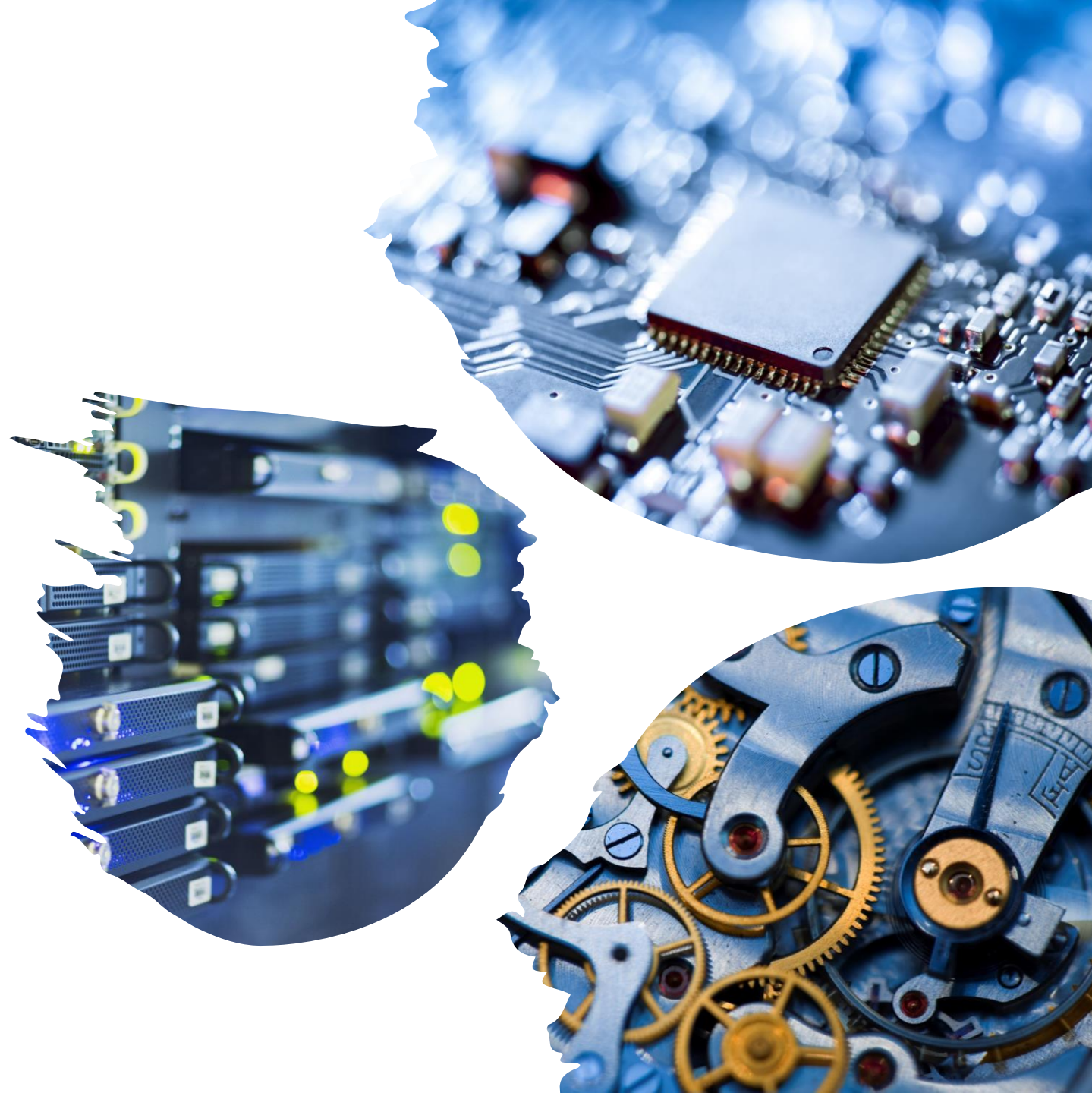
Using either graphical or mathematical method, show there is no diminishing returns

Overcoming the odds ???



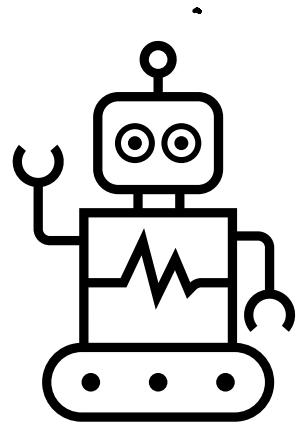
ECONOMIC GROWTH AND ROLE OF TECHNOLOGY

Overcoming the odds



3. Achieving Growth and role of technology

- $Y = A F(K,L)$, e.g. $Y = A(K)^{1/2}(L)^{1/2}$
- A rise in level of technology is also known as Embodied technical progress
- This means increasing A , increases output (Y) with no change in the level of inputs (K and L)



A	K	L	Y	Y/L
1	4	4	4	1
2	4	4	8	2
3	4	4	12	3
4	4	4	16	4
5	4	4	20	5

Seems easy mathematically but impossible in the real world

Both Y and Y/L rise at a constant rate

3.1 Technology

- **Information Technology** (computers, digital technologies, telephony, and the Internet)
- Video Link : Block chain and trade : <https://www.youtube.com/watch?v=b5cP6F4hp-A&feature=youtu.be>
- **Transportation Technology** (Jet aircraft 800 to 1100 KM per hour – Boom Supersonic 1,687mph new York to London 3 hours and 15 minutes) - containerised shipping
- **Manufacturing Technology** (Computer-aided design of products (CAD), robotics, and IT-managed production lines)

Industrial Revolution Timeline



Mid-1700s to 1830

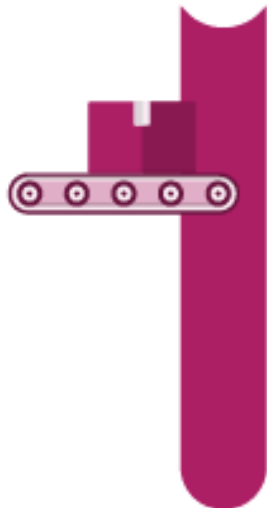
1st industrial revolution

Mostly in Britain

Inventions:

Spinning jenny, water wheel, steam engine, cotton gin, concrete, steam locomotive, steamboat and steamship, modern roads, batteries

Observed that the benefits of these revolution is kept within the country(s) and thus growth and development was uneven in different parts of the world



Mid-1800s to mid-1900s

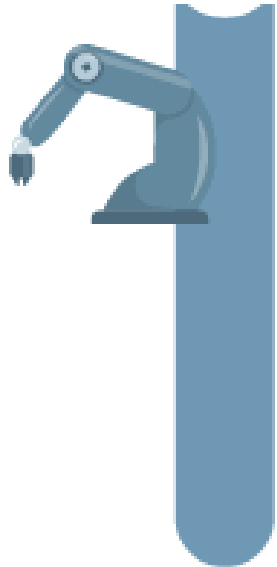
2nd industrial revolution

Europe, North America and Japan

Inventions:

Assembly-line manufacturing, electric rail, plastics, incandescent lamp (lightbulb), telephone, telegraph, internal combustion engine, automobile, typewriter, skyscrapers, motion pictures

Industrial Revolution



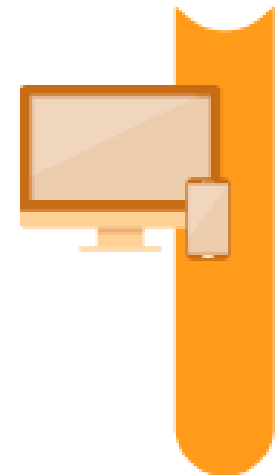
1960s to 2000s

3rd industrial revolution

Everywhere

Inventions:

Computer programming, machine automation, internet, 3D printing, robotics, Internet of Things, nuclear energy, microprocessor, transistor, space research, biotechnology



Now

4th industrial revolution

Everywhere

Inventions:

Machine learning, artificial intelligence, Big Data (data-based decision making), cloud computing, predictive programming

..Will this make a difference especially for the slow growers to catch up? But will this then make the Giants uneasy ???

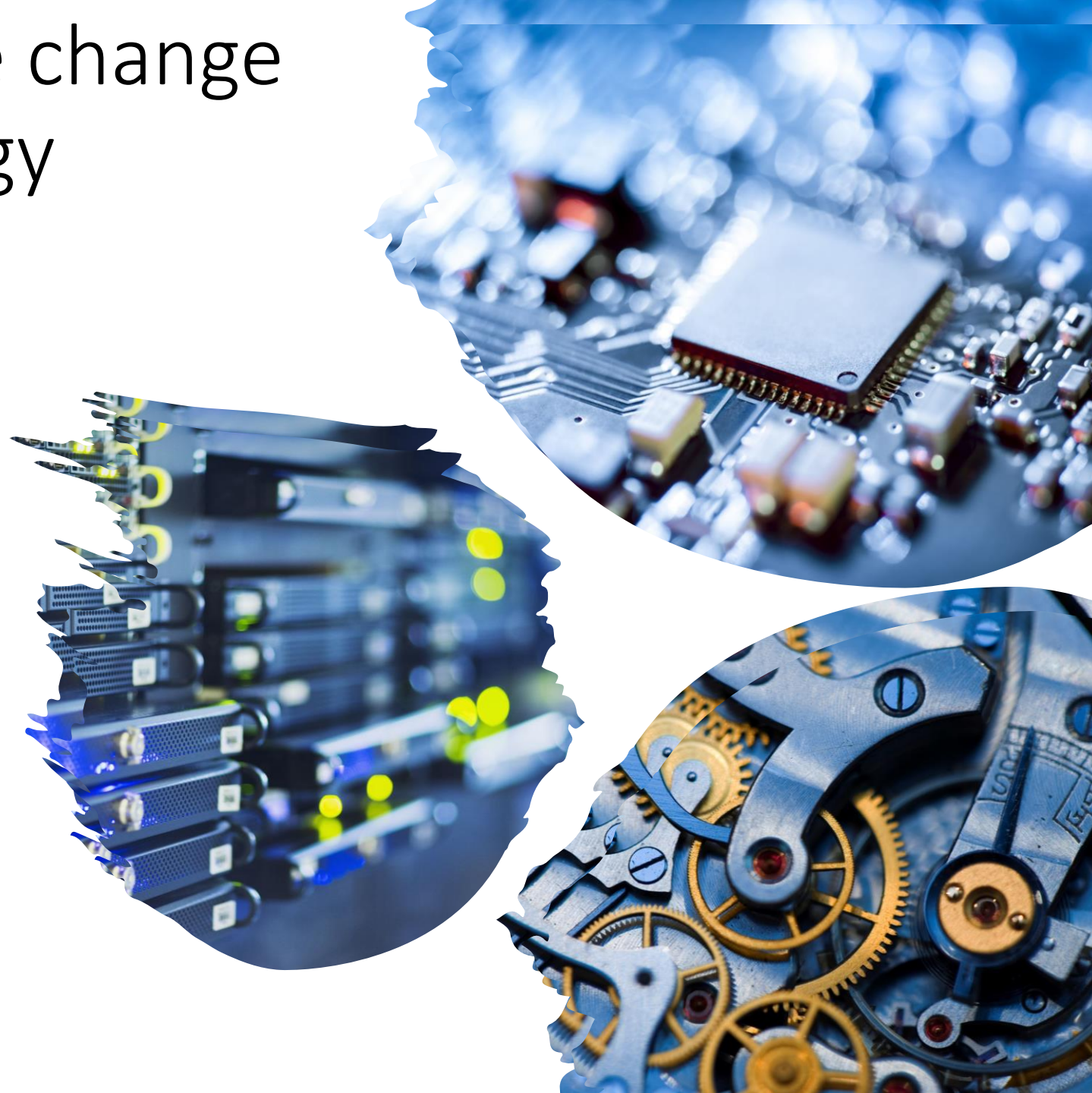
Video links

[CNBC explained](#)

<https://youtu.be/kpW9JcWxKq0>

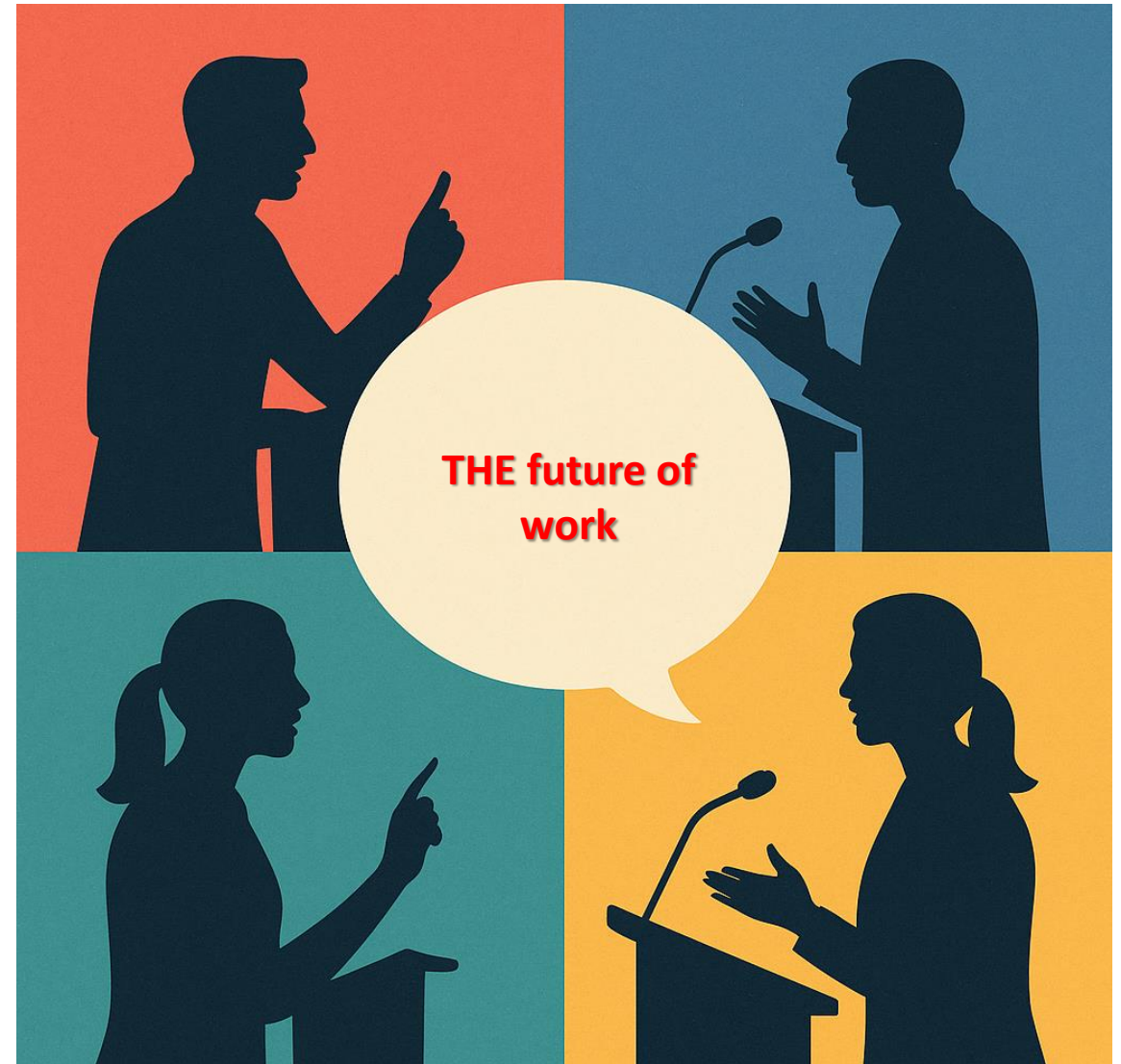
1.2.6 Measuring the change in level of Technology

- **TOTAL FACTOR PRODUCTIVITY (TFP)**
- captures everything that contributes to output growth beyond the easily measurable inputs of capital (K) and labor (L).
- But in some cases when TFP Data is not available, we use other proxy indicators : R&D Expenditure, ICT growth, HCI index



The big debate : Increasing A

- Our simple computation previously assumes both K and L to stay constant while the level of A rises.
- This is favourable as this slows down diminishing returns and increases labour productivity (Y/L)
- But in the long run with advanced technological progress, what will be the **impact on the level of K and L**
- And also. In lecture one, we do have another input known as **human capital (H)**
- $Y = A F(K, L, N, H)$
- So how will this pursuit of increase the level of technology have an impact on the various inputs ?



- Computing and interpreting economic growth rates
- Role of K , L (inputs) to increase output (Y)
- Role of diminishing returns
- Solow's growth model and role of K
- Role of technology

We will examine the specific roles of the variables in determining the level of Real GDP using single and/or multiple regression in tutorial 2

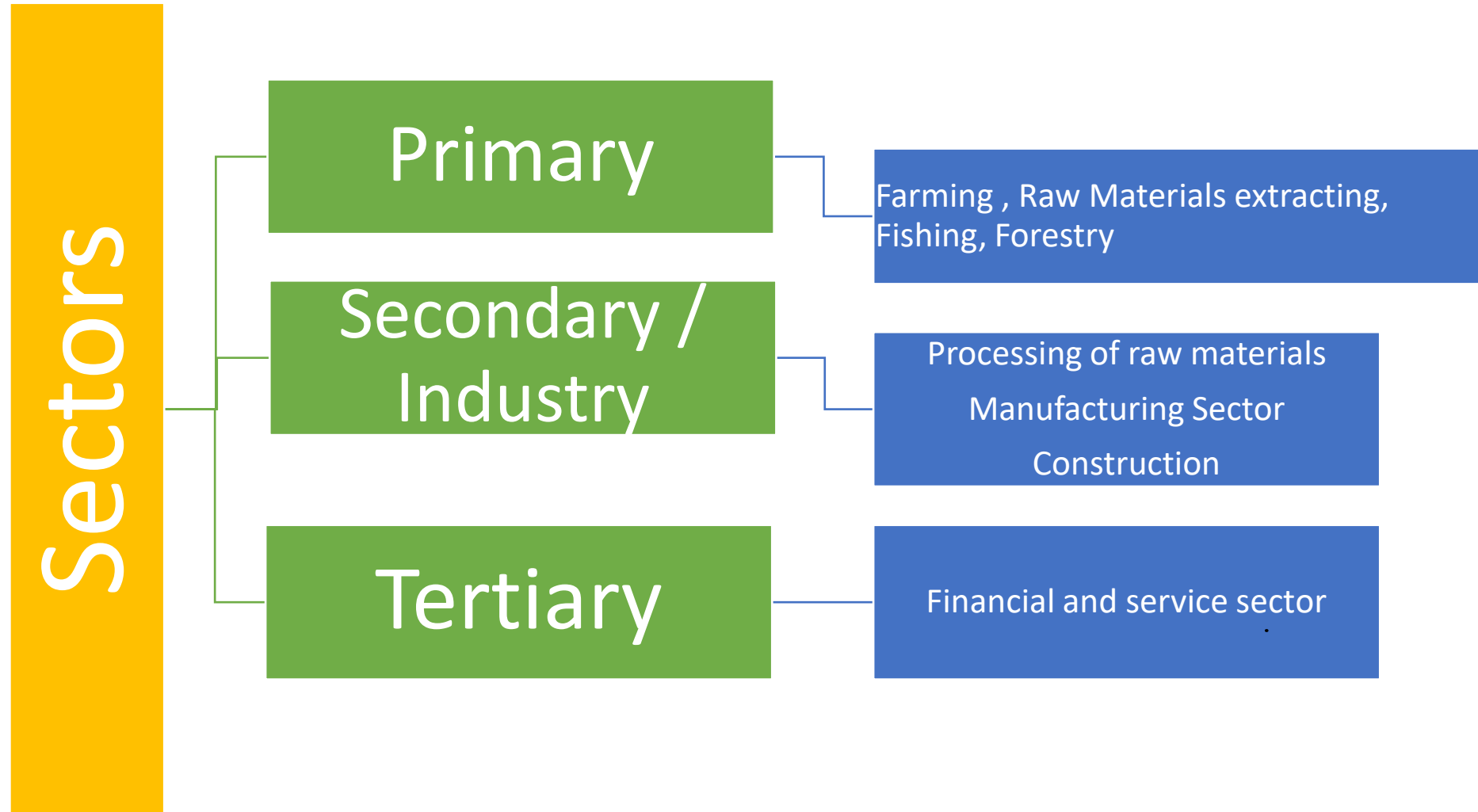


Part 2

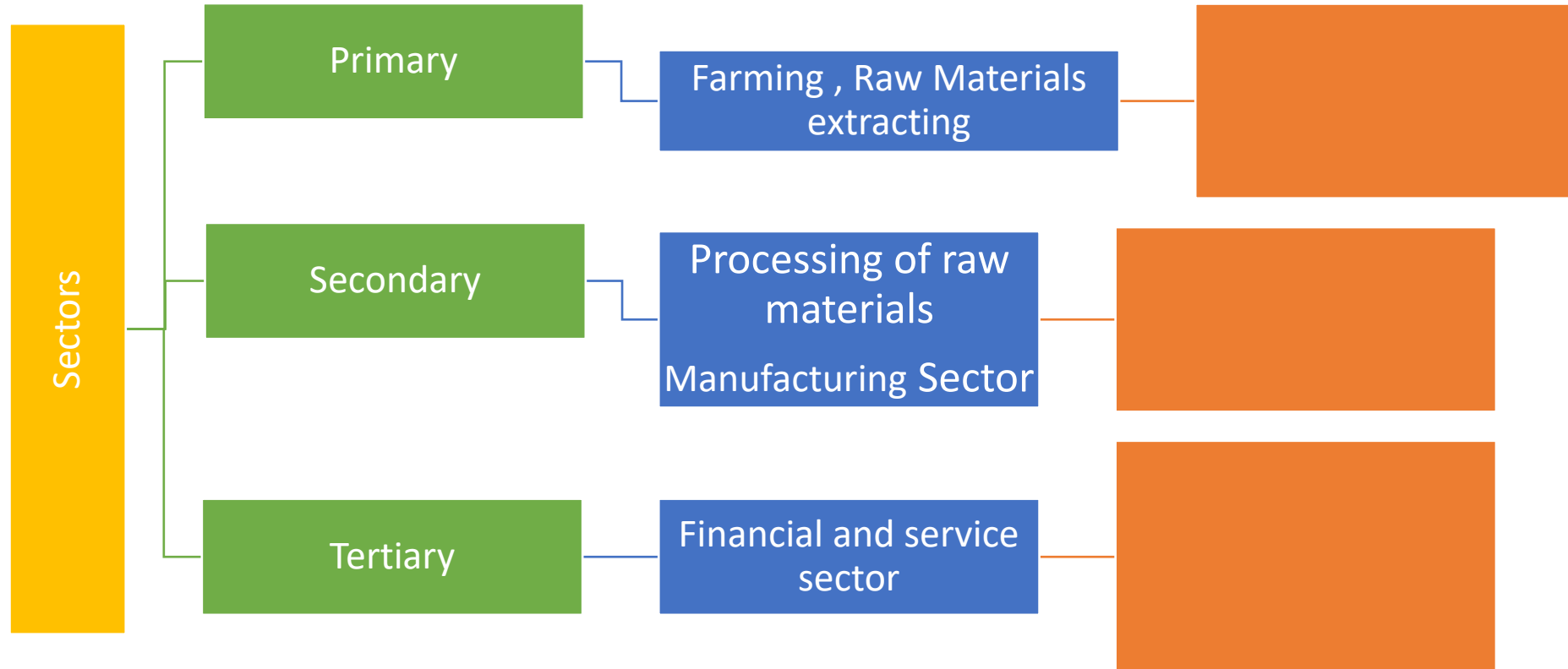
Economics Sectoral Growth



4. Economic Sectors



3.1.1 Exercise



Exercise 1:

In which sector will you classify for the following production ?

Rice

Insurance

Electric cars

Computer chips

Oil

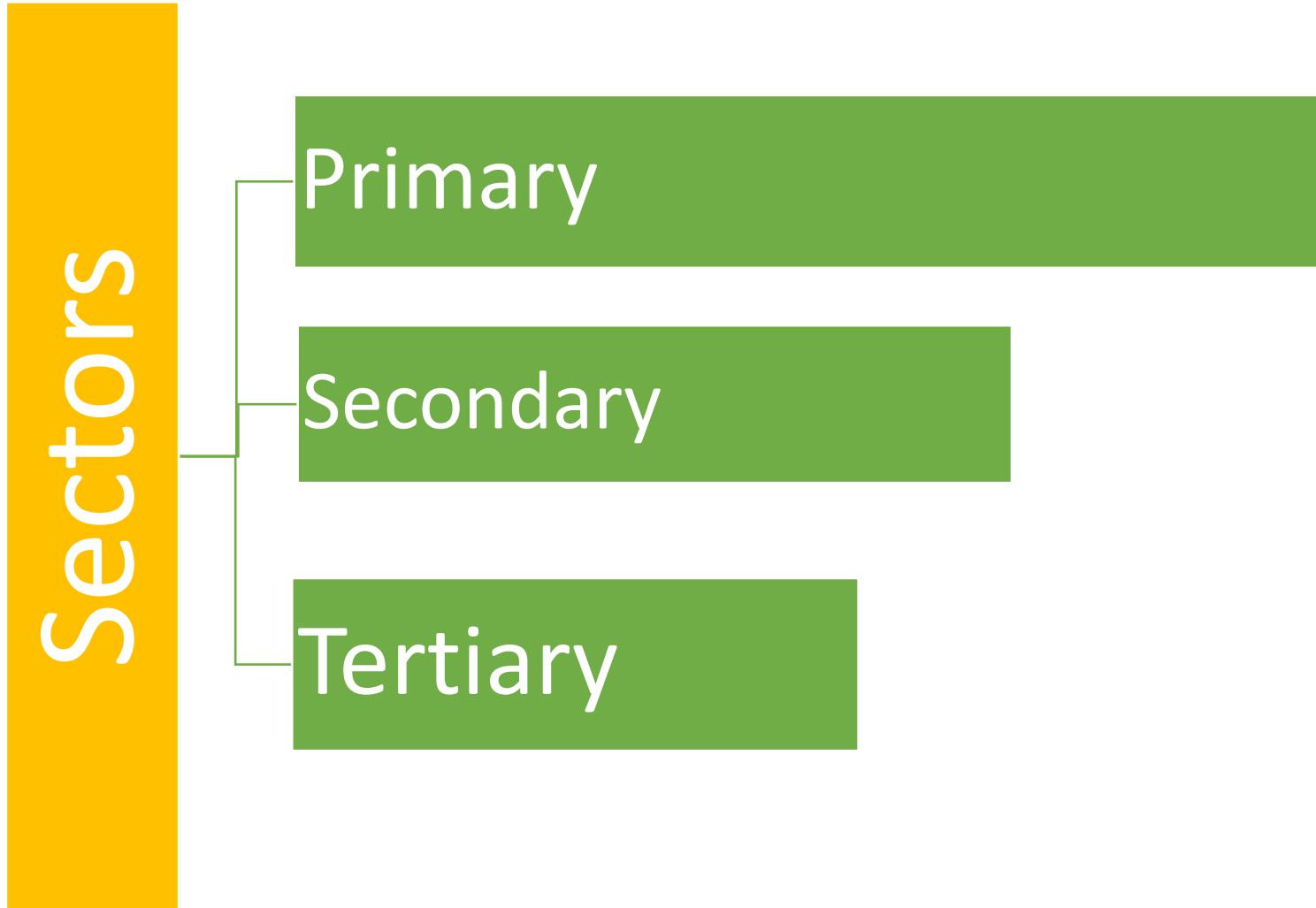
Cotton

Web hosting

Natural Gas

Lab grown corn

3.1.2 Economic Sectors – Exercise



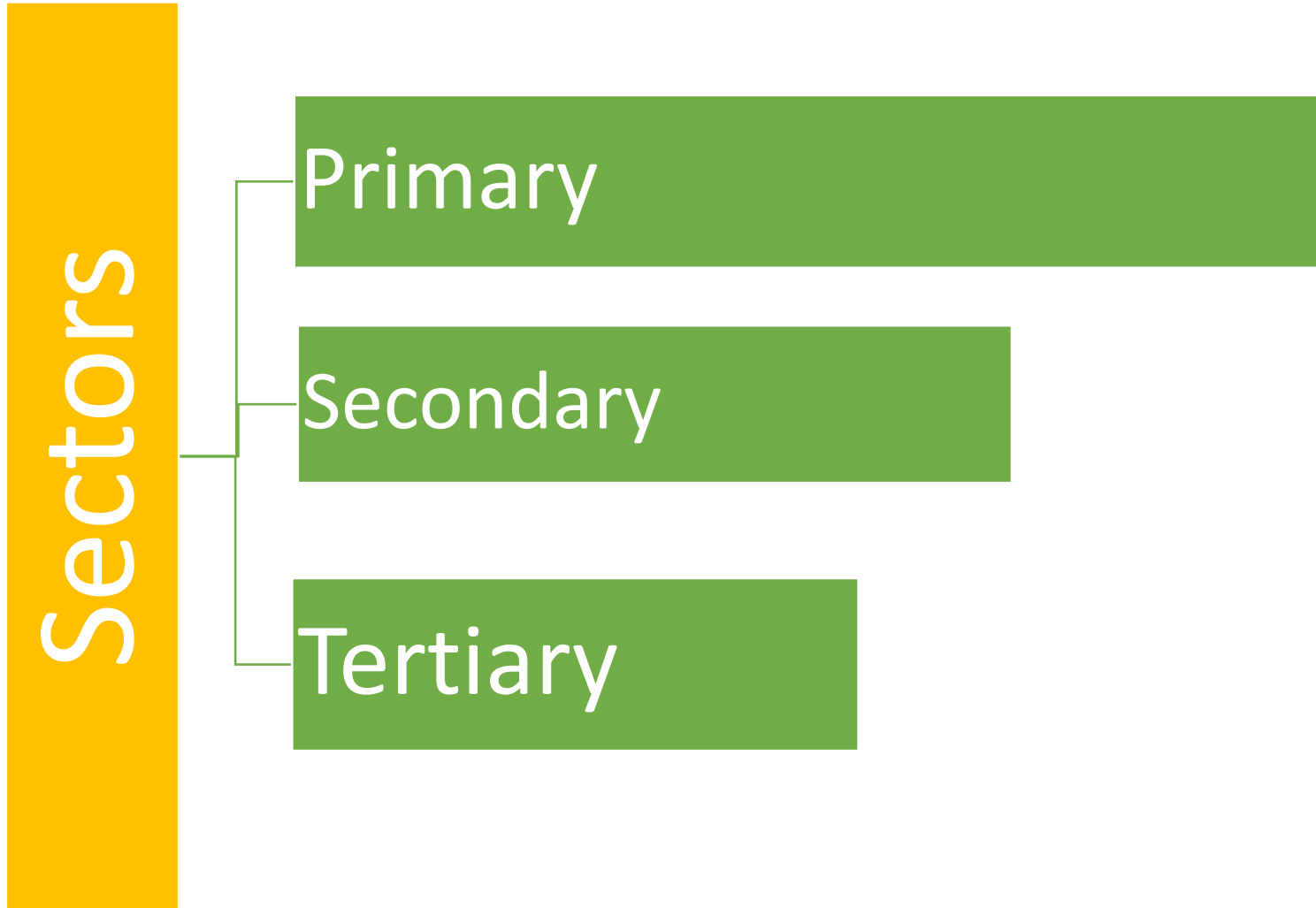
Exercise 2 :

Using the basic Output model

$$Y=AF(K,L,N,H)$$

Indicate for each sector, which input(s) will you consider to be a key contributor to the sector's output

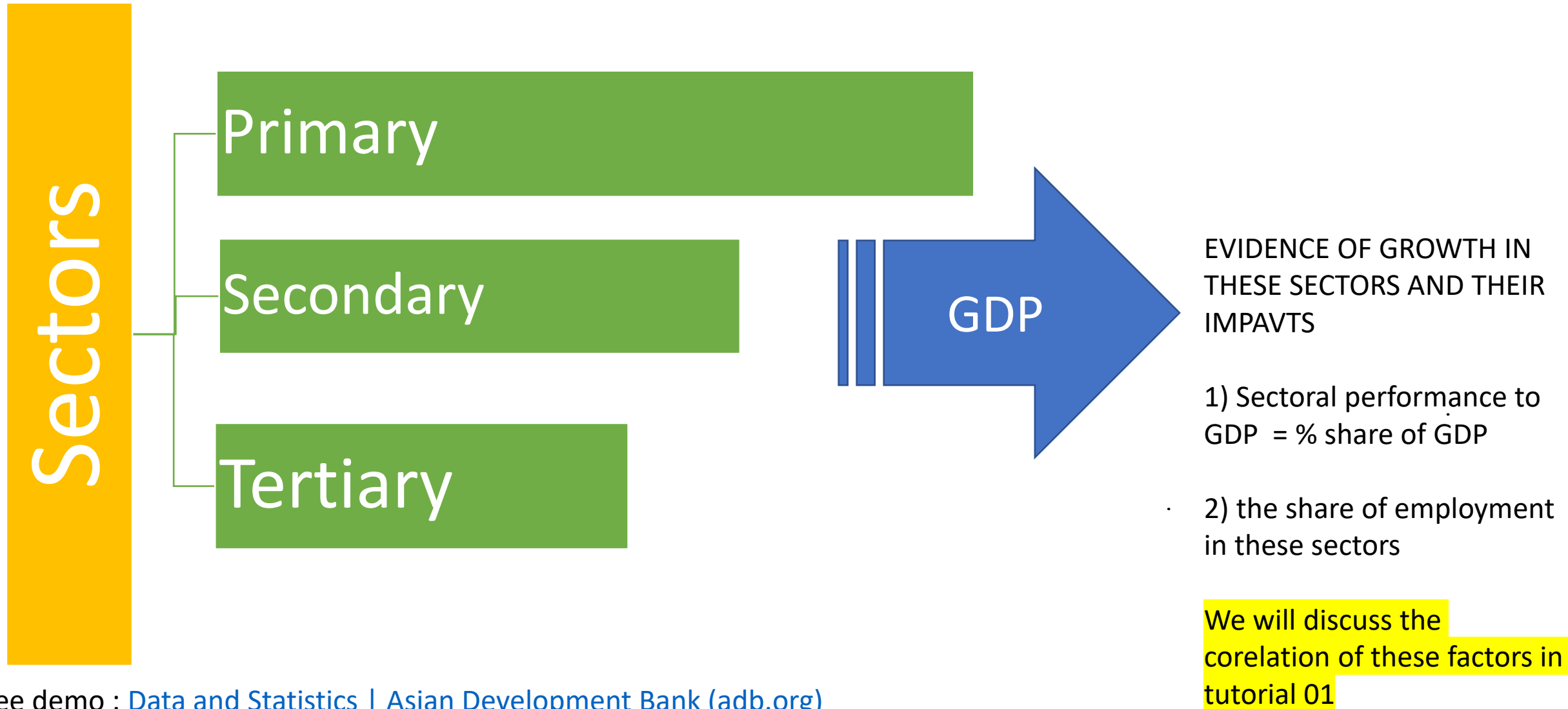
3.1.2 Economic Sectors – Exercise



Exercise 3 :

Using either one of the growth models covered in Section 2 , rate in descending order (scale of 1-3) the sector will bring about more growth to the economy

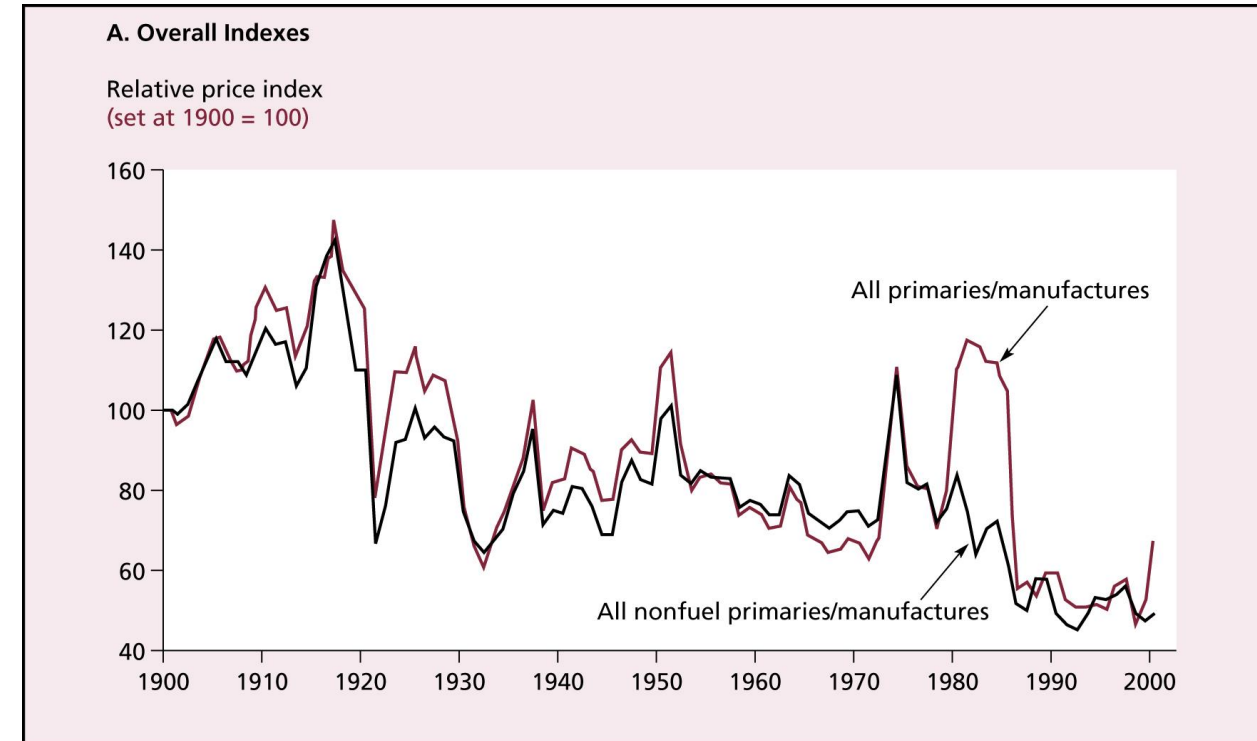
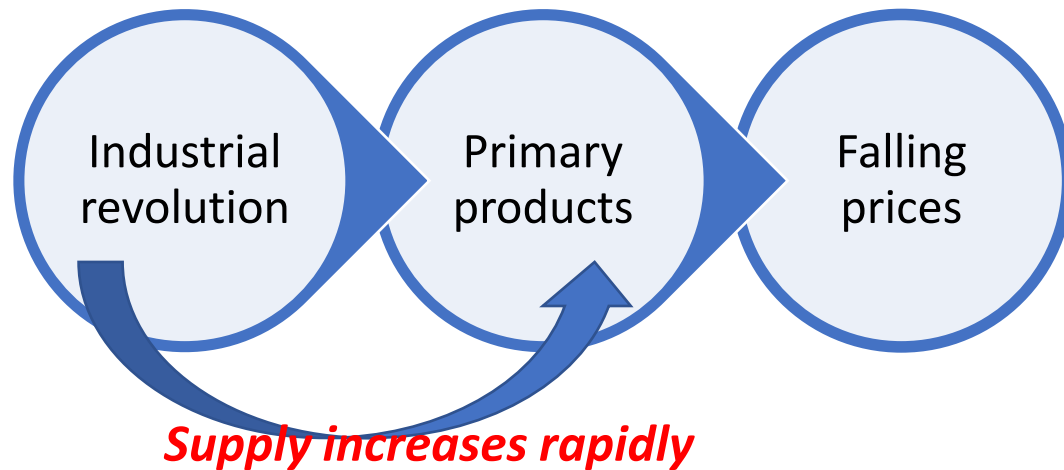
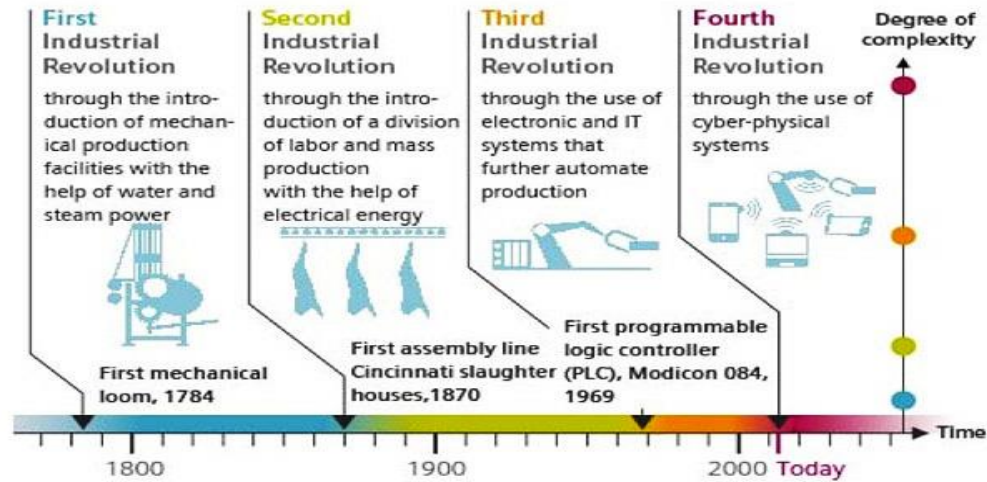
3.1.2 Economic Sectors – Evidence

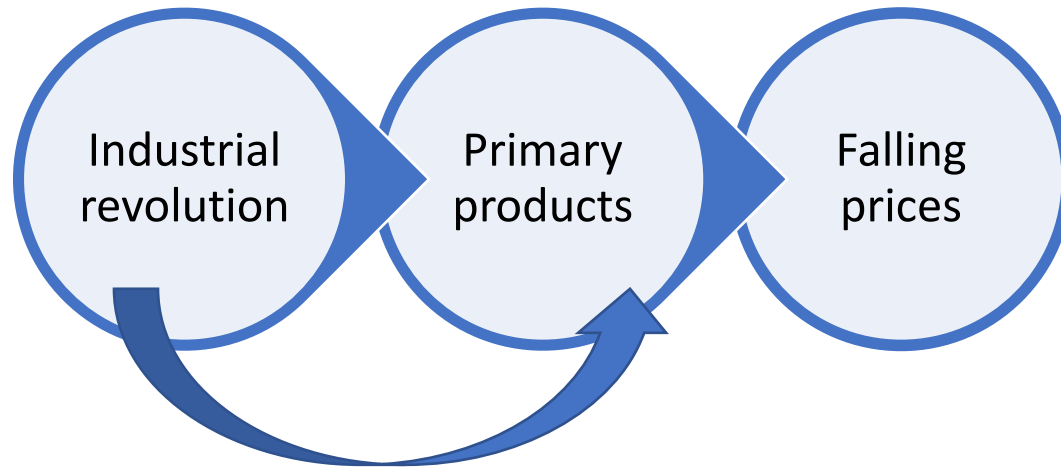
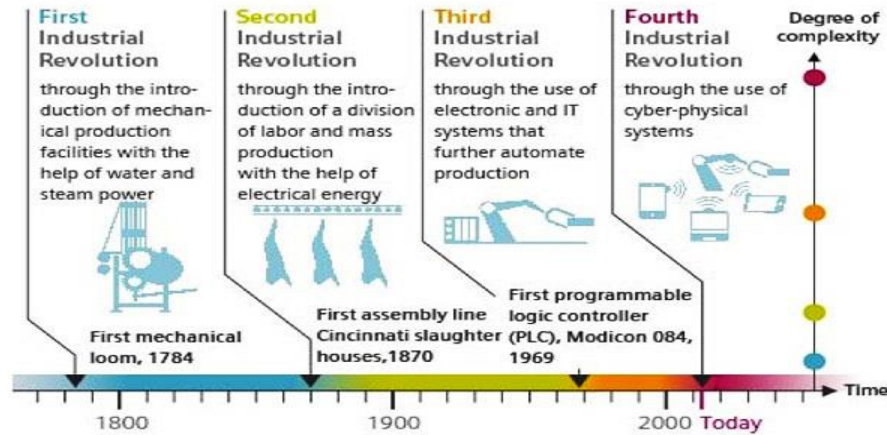


See demo : [Data and Statistics](https://data.adb.org/) | [Asian Development Bank \(adb.org\)](https://adb.org/)

4.2 The primary sector

Given the falling prices and more rapid production, demand will go up; so isn't that good?





Supply increases rapidly

Consider if total revenue = **Price X Quantity**

A 10% fall in the price with a **MUCH SMALLER rise** of 2% rise in qty

What happens to total revenue ? **FALLS**



Other key observations

Consider the nature of the good:

1. **Unstable supply**
2. **Perishable**
3. **Complex storage system and transportation**

Implications on trade : Intra-national and international trade

4.3 Economic Sectors : The Secondary or Manufacturing Sector

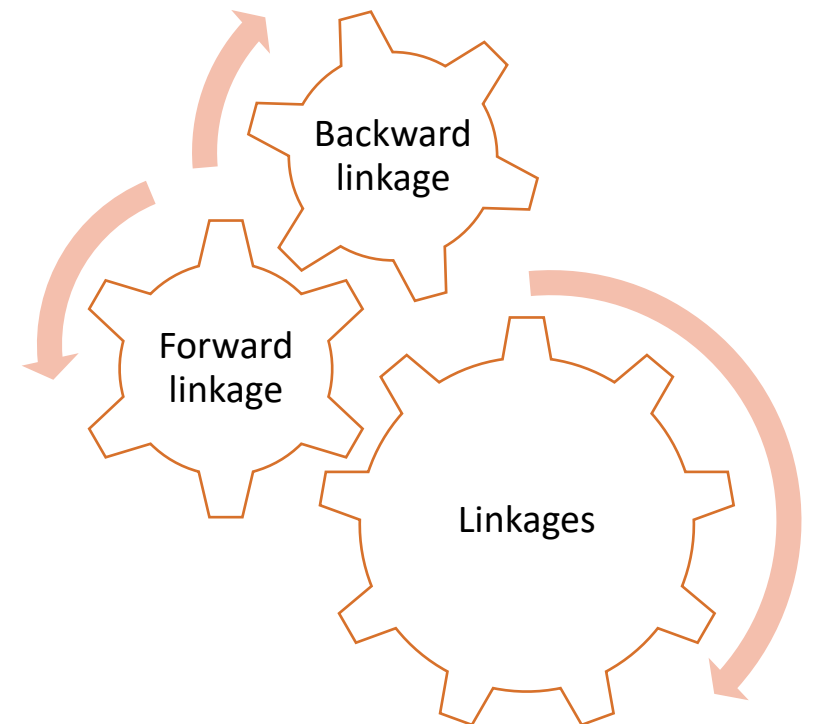


Manufacturing is the **only sector** that in the long-run subject to

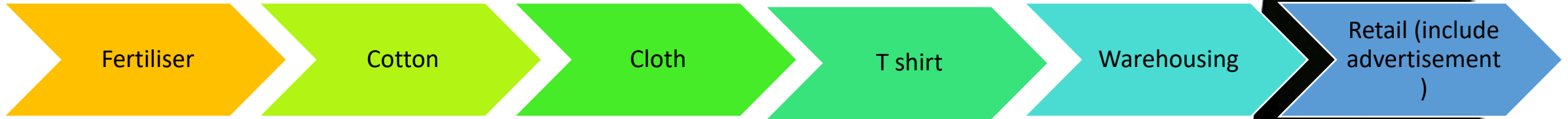
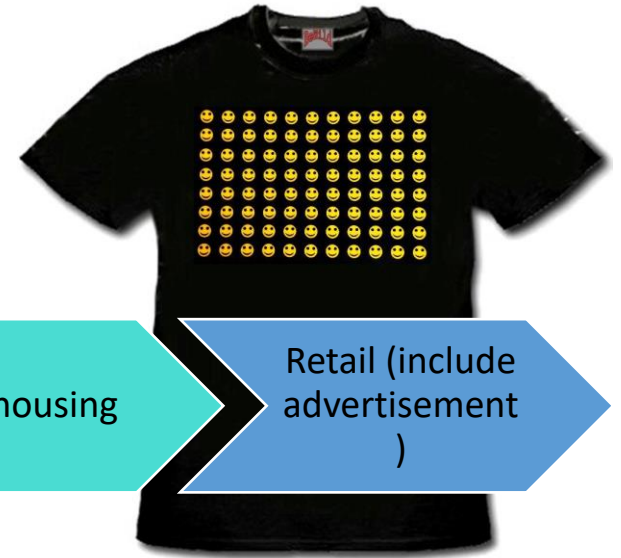
- + rising productivity as output expands.
- + inter-related branches of activity.
- + produces capital goods used in a range of sectors
- + has relatively more linkages
- + engine of growth

4.3.1 Manufacturing and role of Linkages

- The activities with the highest linkages have the potential for generating externalities (i.e. spill over effects)
- Backward linkages from a particular industry to its suppliers
- Forward linkages from an industry to its users



4.3.1 Forward & Backward Linkages



Product	Number of backward linkages
Cotton	
Cloth	
T shirt	

So why will
occurrence of more
backward linkages
matter more to
economic growth ?

A broad summary

Mainly by poor
developing
economies

Primary goods

Mainly by rich developed
economies with funding

Manufacturing
goods

Smaller % of K and A

Unstable supply

Low GDP and growth

Higher % of K and A

Stable supply

Linkages

Engine of growth



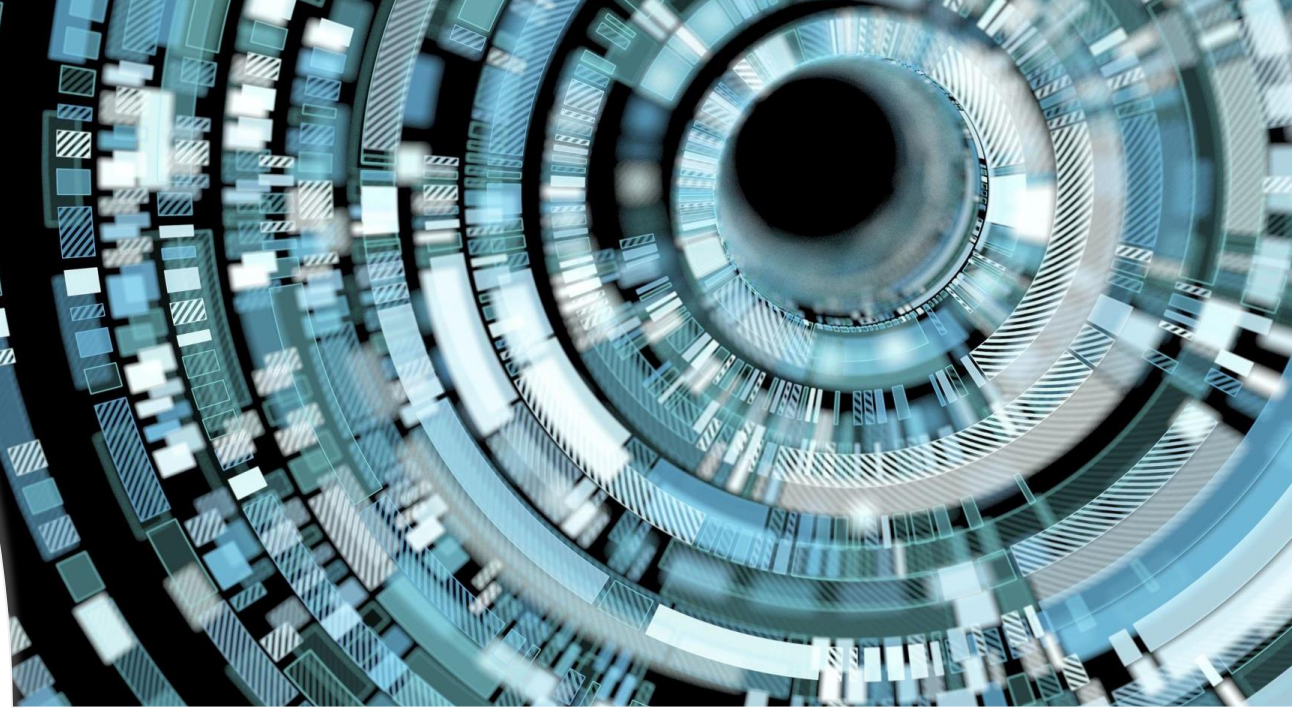
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5. The other input : Natural resources (N)

The Extraction Sector

- At one glance ,we may conveniently classify this as the primary sector
- But a careful examination of some primary raw materials such as Crude oil, Rare Earth Elements
- They are different from the other primary products such as crops as discussed earlier
- More than often the processing of the minerals are classified in industry sector



5. The other input : Natural resources (N)

The Extraction Sector

- Requires extensive and sophisticated processing : manufacturing sector
- Rising global demand
- GDP contribution ?? – often recorded separately
- The next race on Rare Earth Elements ;

Link : https://youtu.be/gyH_PvLZoD0



5. DEPENDENCY THEORY :



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Poor
developing
economies

Export Cheaper
primary goods

Import the pricier
manufactured
goods

Rich developed
economies

Import cheaper raw materials
to produce more pricey
manufactured goods

Earn less but pay more

Thus many economies take on the path to develop the secondary sector : Path of Industrialisation : Topic 3



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